Social Media Engagement Analytics Using Python

One-Page Project Summary

Project Overview

The **Social Media Engagement Analytics** project analyzes social media data using Python to understand content performance, user behavior, engagement patterns, device usage, and sentiment. The dataset contains **5,000 records** with information such as post type, category, likes, comments, shares, impressions, watch time, followers, country, age, gender, device type, sentiment, and engagement rate.

Tools & Libraries Used

Python, Pandas, NumPy, Matplotlib, Seaborn, and Plotly

Data Cleaning & Preparation

The dataset was checked for missing values, duplicates, incorrect data types, and unrealistic values. Missing numerical values were handled using the **median**, while categorical missing values were filled using the **mode**. Gender and sentiment labels were standardized, and the posted_at column was converted to datetime format. New features such as **hashtag count, engagement score, log impressions, age group, and posting hour** were created for further analysis.

Exploratory & Statistical Analysis

Pandas and NumPy were used to explore the dataset using summary statistics, categorical distributions, correlation analysis, and groupby() operations. Statistical measures such as **mean, median, mode, standard deviation, variance, percentiles, skewness, and kurtosis** were calculated for important engagement metrics.

Data Visualization

More than **8 visualizations** were created using Matplotlib and Seaborn, including scatter, line, bar, pie, histogram, box, count, violin, pair, heatmap, and swarm plots. An **interactive Plotly scatter chart** was also created to explore the relationship between impressions, likes, post types, and engagement.

Key Findings

- **Video** was the highest-engagement post type with an average engagement score of **12,664.59**.
- **Music** was the best-performing content category with an average engagement score of **12,788.55**.

- **Brazil** had the highest average engagement rate at approximately **1.54**.
- Users aged **31–40** recorded the highest average engagement score of **12,938.50**.
- Verified accounts showed higher average engagement rates and impressions, while non-verified accounts had a slightly higher average engagement score.
- **Mobile** users had the highest average watch time at **4,087.83 seconds**.
- **Negative-sentiment posts** had the highest average engagement score and engagement rate, while positive posts had slightly higher average impressions.
- Best posting time could not be reliably determined because the dataset did not contain varying time-of-day information.

Conclusion

The project successfully demonstrates how Python can be used to **clean, transform, analyze, and visualize social media data**. The findings show that engagement varies across content type, category, country, age group, verification status, device type, and sentiment. These insights can support better content planning and data-driven social media strategies.